

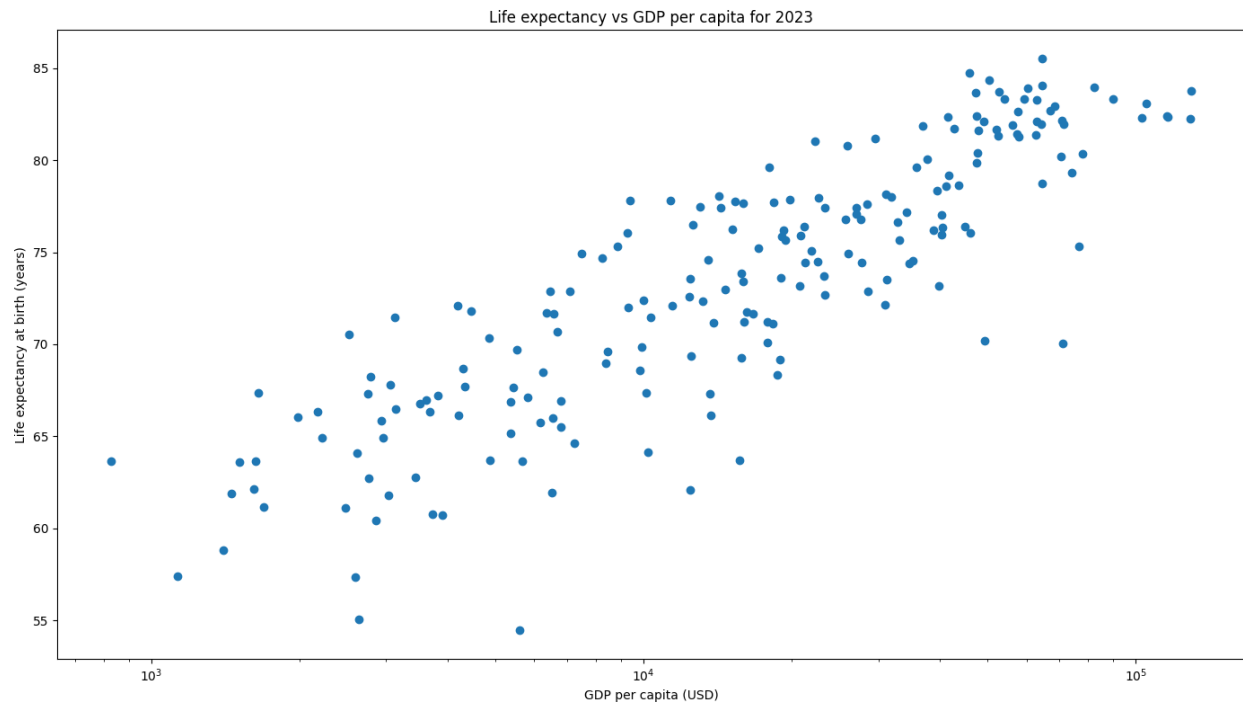
Economic growth and Life expectancy in the world

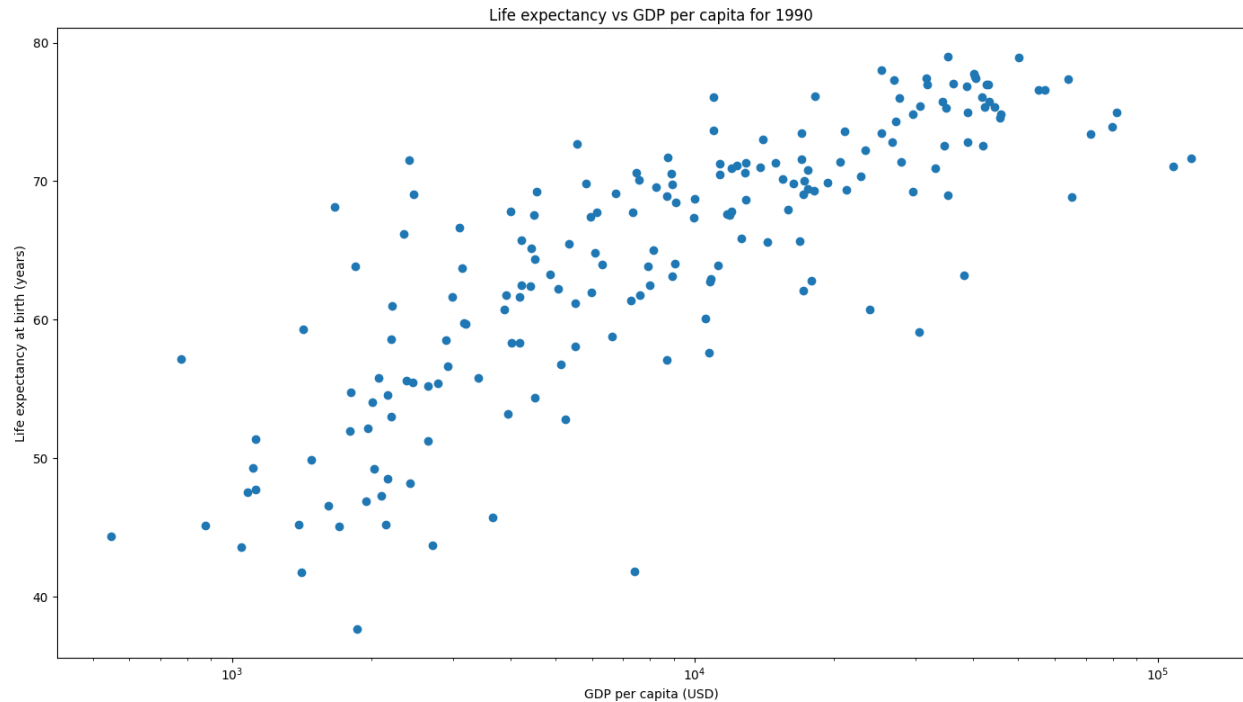
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After advanced research of the provided data about Economic growth, meaning the quantity or quality of the many goods and services that people produce and Life expectancy throughout the countries in the world, all features were taken into account in order to retrieve meaningful information about the countries and their population.

In this paper, results of the research performed will be presented. Precise data analysis can be found in the Jupyter notebook describing the process for the decisions taken.

- a. Write a Python program that draws a scatter plot of GDP per capita vs life expectancy. State any assumptions and motivate decisions that you make when selecting data to be plotted, and in combining data



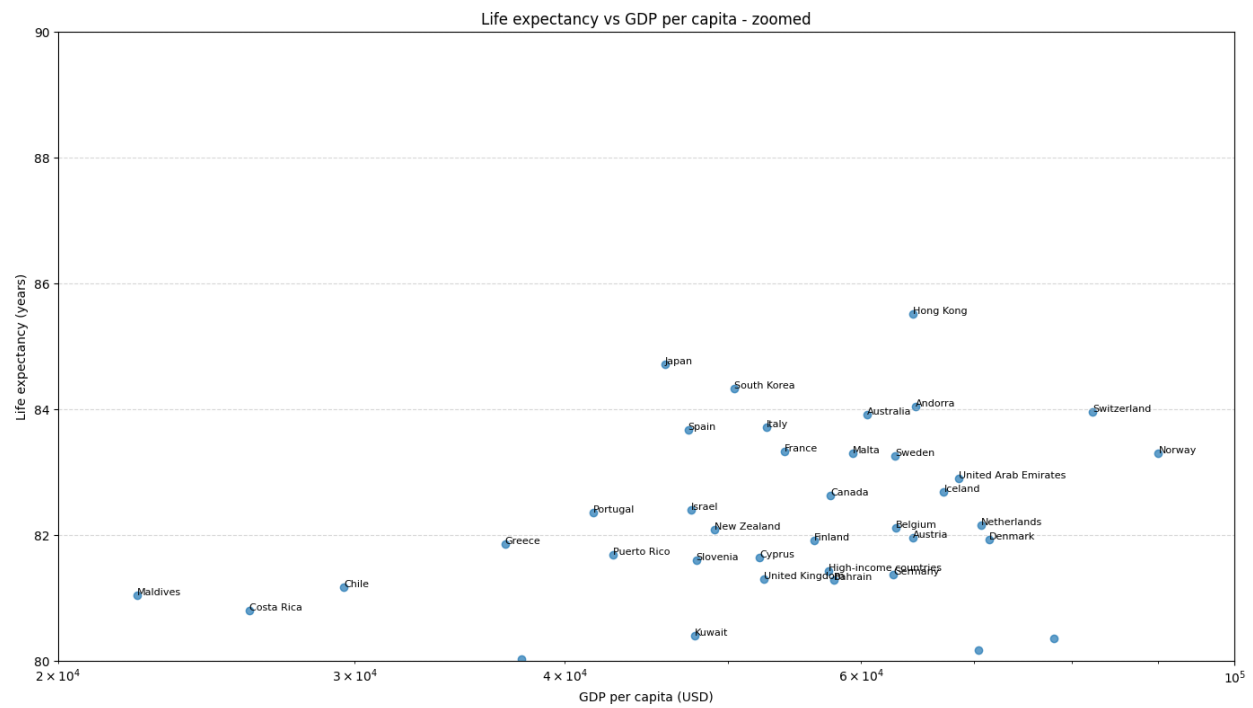
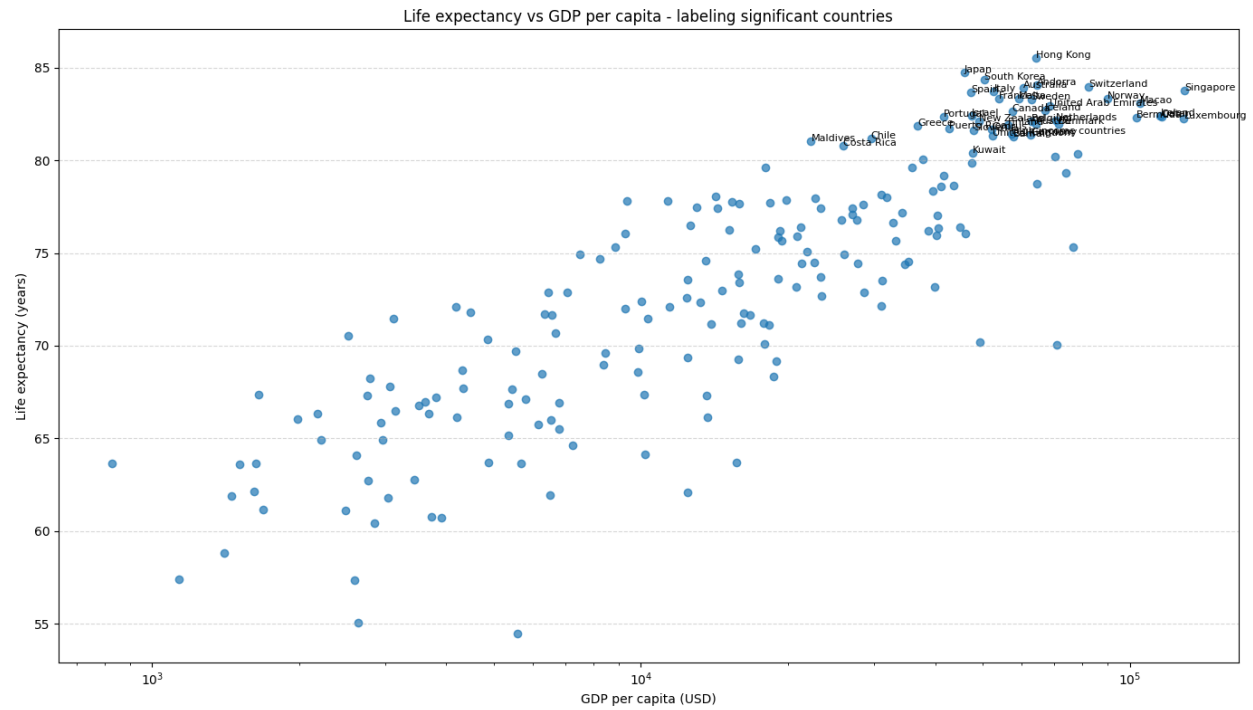


A strong correlation between the GDP per capita and the Life expectancy of a person can be seen. The more the GDP per capita is, the more is the average life expectancy of a person in the country.

We can see that the correlation between the two features is also visible in 1990 although there are some more distractions in the data.

b. Which countries have a life expectancy higher than one standard deviation above the mean?

41 countries are with so high life expectancy as it is visible in the list below. The countries with life expectancy higher than one standard deviation above the mean are mostly with very high GDP per capita (they are situated in the upper right corner of the scatter plot)



	Entity	Life_expectancy	GDP_per_capita
0	Hong Kong	85.5106	64428.730
1	Japan	84.7123	45858.660
2	South Korea	84.3288	50414.207

3	Andorra	84.0406	64631.297
4	Switzerland	83.9536	82302.050
5	Australia	83.9228	60461.156
6	Singapore	83.7362	129555.240
7	Italy	83.7157	52725.746
8	Spain	83.6703	47341.170
9	France	83.3253	54017.875
10	Norway	83.3078	90085.555
11	Malta	83.2989	59293.297
12	Sweden	83.2624	62845.484
13	Macao	83.0749	104961.920
14	United Arab Emirates	82.9091	68577.540
15	Iceland	82.6911	67177.200
16	Canada	82.6302	57517.440
17	Ireland	82.4118	115505.336
18	Israel	82.4080	47527.152
19	Qatar	82.3680	116159.140
20	Portugal	82.3599	41571.020
21	Bermuda	82.3089	103052.350
22	Luxembourg	82.2294	129021.650
23	Netherlands	82.1576	70673.970
24	Belgium	82.1150	62920.645
25	New Zealand	82.0880	49083.508
26	Austria	81.9559	64394.090
27	Denmark	81.9328	71454.520
28	Finland	81.9102	56246.440
29	Greece	81.8568	36854.504
30	Puerto Rico	81.6902	42738.645
31	Cyprus	81.6480	52190.875
32	Slovenia	81.6034	47868.387
33	High-income countries	81.4353	57349.350
34	Germany	81.3777	62686.540
35	United Kingdom	81.3015	52502.914
36	Bahrain	81.2835	57819.906
37	Chile	81.1667	29563.678
38	Maldives	81.0412	22286.980
39	Costa Rica	80.7995	25979.629
40	Kuwait	80.4047	47777.402

c. Which countries have high life expectancy but have low GDP? Motivate how you have chosen to define “high” and “low”.

We can define Low and High depending on the situation and the needs of the experiment. It can be defined just above and below the mean value. In the first experiment, we checked for the countries which have GDP per capita in the first quantile, meaning below 25%, but such don't exist.

With GDP below the mean value two countries exist – Maldives and Costa Rica.

	Country	Life_expectancy	GDP_per_capita
113	Maldives	81.0412	22286.980
42	Costa Rica	80.7995	25979.629

In conclusion, we decreased the life expectancy threshold to the mean value.

	Country	Life_expectancy	GDP_per_capita
190	Ukraine	73.4225	15916.9580
18	Belize	73.5656	12455.2960
174	Suriname	73.6310	19043.7050
52	Dominican Republic	73.7201	23282.4900
143	Paraguay	73.8443	15826.0880
11	Azerbaijan	74.4286	21310.7380
67	Georgia	74.4965	22590.5300
198	Vietnam	74.5883	13545.9350
14	Bangladesh	74.6723	8242.3980
118	Mauritius	74.9263	26060.7320
132	Nicaragua	74.9468	7496.5293
119	Mexico	75.0690	21905.1780
72	Grenada	75.2052	17128.6970
124	Morocco	75.3128	8868.7120
7	Armenia	75.6827	19402.7520
25	Brazil	75.8482	19079.8120
194	Upper-middle-income countries	75.8812	20882.4120
33	Cape Verde	76.0587	9280.9795
15	Barbados	76.1782	19224.4630
2	Algeria	76.2610	15159.3240
180	Thailand	76.4115	21191.4530
184	Tunisia	76.5077	12622.9600
160	Serbia	76.7693	25740.2030
54	Ecuador	77.3924	14343.0190

135	North Macedonia	77.3947	23343.6020
172	Sri Lanka	77.4828	13024.9400
85	Iran	77.6545	15912.0300
39	Colombia	77.7246	18382.9980
144	Peru	77.7401	15327.5820
92	Jordan	77.8145	9381.3545
101	Lebanon	77.8170	11330.2690
23	Bosnia and Herzegovina	77.8504	19803.8460
38	China	77.9529	22687.0760
96	Kosovo	78.0326	14240.2030
1	Albania	79.6019	17991.0160
42	Costa Rica	80.7995	25979.6290
113	Maldives	81.0412	22286.9800

In this example much more countries exist, however, the requirements are just life expectancy above the mean value and GDP per capita below the mean value which involves countries right in the middle. This last result is not giving correct enough information on the subject.

d. Does every strong economy (normally indicated by GDP) have high life expectancy?

	Country	GDP_per_capita	Life_expectancy
71	Greenland	71037.54	70.0550
76	Guyana	49312.36	70.1802

Although only two countries complete the requirements, this is proving that the high economy is not ensuring high life expectancy. Things like environment and religion could be fundamental for defining life expectancy.

e. Related to question d, what would happen if you use GDP per capita as an indicator of strong economy? Explain the results you obtained, and discuss any insights you get from comparing the results of d and e

Correlation between GDP per capita and life expectancy: 0.744

In conclusion, we can see that the correlation is almost 0.75 which is in the highest 25%, however it is not enough to conclude that high GDP per capita results in a strong economy.